# Twin Layer: Physical Simulation & Adversary Specifications

This reference documents the physical formulas, diurnal kinetic curves, incident engine dynamics, and adversary state machines powering the simulation engine in `twin/simulation.py`.

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## 1. Simulation Engine Principles & Dual-Clock System

1. **Dual-Clock Coordination:**
2. **Simulation Time:** Strictly deterministic RFC 3339 UTC timestamps ( `YYYY-MM-DDTHH:MM:SSZ` ).
3. **Wall-Clock Pacing:** Cycle pacing runs strictly at 10 Hz (100 ms per wall cycle) via `SimulationClock.pace_wall_clock()`. Wall-clock time is never leaked into Kafka payloads, SQLite truth readings, or MCP responses.
4. **Simulation Speed Acceleration ( `SIM_SPEED` ):**
5. Configurable via environment variable `SIM_SPEED` (default `1.0` , allowable range `0.1` to `60.0` ).
6. Each simulation tick advances simulation time by $\Delta t = 0.1 \times \text{SIM_SPEED}$ simulation seconds.
7. All physical differential equations, noise terms, and exponential lags are written per simulation second, ensuring physical trajectory invariance regardless of `SIM_SPEED` .
8. **Euler-Maruyama Stochastic Integration & Lags:**
9. Noise terms scale with $\sqrt{\Delta t}$ in accordance with the Euler-Maruyama scheme: $\sigma \sqrt{\Delta t} \cdot \mathcal{N}(0, 1)$.
10. Dynamic inertia follows first-order exponential lag: $\alpha = 1 - \exp(-\Delta t / \tau)$, where $\tau$ is the characteristic time constant in simulation seconds.
11. **Dedicated Physics PRNG ( `physics_rng` ):**
12. Physics updates, camera variety factors, and stochastic Poisson arrival processes derive exclusively from a dedicated instance of `random.Random(SIM_SEED)` .
13. Network interactions, mock clients, and API requests do not advance the physics RNG, guaranteeing bit-for-bit trajectory determinism across identical runs.
14. **Municipal Zones & Geographic Coordinates:**
15. The city comprises 42 IoT devices distributed evenly (7 devices per zone) across 6 municipal zones.
16. Zone-2 ("Port & Coastal Harbor") is positioned at the residential center of **La Goulette** with centroid coordinates `(36.8185, 10.3050)` .
17. The 7 devices in each zone follow the deterministic geographic offset formula:
    $$\text{lat} = \text{round}(z_{\text{lat}} + (\text{idx} \times 0.0012) - 0.0036, 6)$$
    $$\text{lon} = \text{round}(z_{\text{lon}} + (\text{idx} \times 0.0015) - 0.0045, 6)$$

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## 2. Sensor Kinetics & Physical Formulations

The digital twin models 42 IoT devices across 5 municipal categories:

### A. Traffic Camera ( `traffic_camera` )

- **Metric:** Vehicle passage throughput in vehicles per minute ($V$).
- **Operating Bounds:** $0.0 \le V \le 200.0\text{ veh/min}$.
- *Diurnal Target Demand ($V^(t)$):*
  *For simulation hour $t \in [0.0, 24.0)$:*
  $$V^(t) = 25.0 + 95.0 \cdot \exp\left( -\frac{(t - 8.5)^2}{2 \cdot 1.2^2} \right) + 110.0 \cdot \exp\left( -\frac{(t - 17.5)^2}{2 \cdot 1.5^2} \right)$$

- Morning rush hour peaks around 08:30 (~120 veh/min).
- Evening rush hour peaks around 17:30 (~135 veh/min).
- Overnight baseline remains below 35 veh/min (~25 veh/min).
- **Per-Camera Variety:** Each traffic camera is assigned a fixed variety multiplier drawn from $[0.85, 1.15]$ using `physics_rng`.
- **Actuation Impact ( `set_signal_plan` ):**
- `normal` : Multiplier $1.0$.
- `green_wave` : Multiplier $1.15$ (clears congestion corridors).
- `red_restriction` : Multiplier $0.55$ (dampens incoming vehicle density).
- `all_red` : Multiplier $0.05$ (emergency cordon stop).
- `adaptive` : Targets the zone-wide mean target across both cameras in the zone.
- **Incident Impact ( `traffic_accident` ):**
- Reduces target throughput to $\times 0.3$ ramped over 60 simulation seconds.
- **Dynamics:** Time constant $\tau = 60\text{ s}$, noise $\sigma = 0.6\sqrt{\Delta t}\text{ veh/min}$. Steady-state spread $\sigma\sqrt{\tau/2} \approx 3.29\text{ veh/min}$.

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## B. Water Level Float ( `water_level` )

- **Metric:** Drainage channel flood level in meters ($H$).
- **Operating Bounds:** $0.0 \le H \le 12.0\text{ meters}$.
- **Differential Equation:**
$$\frac{dH}{dt} = Q_{\text{in}}(t) - k \cdot c \cdot \sqrt{2g \max(0, H - H_0)}$$
- **Constants & Parameters:**
- Gravitational acceleration: $g = 9.81\text{ m/s}^2$.
- Retention threshold: $H_0 = 1.2\text{ meters}$.
- Calibrated discharge coefficient: $c = 0.00089$.
- Inflow rate: $Q_{\text{base}} = 0.003\text{ m/s}$ modulated by a $\pm 10\%$ tidal term with a 12.4-hour period ($44,640\text{ s}$):
$$Q_{\text{in}}(t) = 0.003 \cdot \left(1.0 + 0.10 \cdot \sin\left(\frac{2\pi \cdot t_{\text{sim_sec}}}{44640}\right)\right)$$
- **Actuation Impact ( `set_valve` ):**
- Valve ratio $k$: `open` ($1.0$), `half` ($0.5$), `closed` ($0.0$), `"NN%"` ($\text{NN}/100$).
- Default initial valve position: `"half"` ($k = 0.5$).
- Initial baseline water level: $3.5\text{ meters}$.
- **Calibrated Behaviors:**
- *Nominal Half Valve:* Holds steady between $2.5\text{ m}$ and $4.5\text{ m}$ for 2+ simulation hours.
- *Closed Valve:* Starting from $3.5\text{ m}$, crosses the $8.0\text{ m}$ critical threshold after $15$ to $40$ simulation minutes (~25 min).
- *Open Valve:* Rapid gravity drainage settles near $1.78\text{ m} \approx 1.8\text{ m}$.
- *Rain Storm ($\times 3$ Inflow) with Half Valve:* Crosses $8.0\text{ m}$ within $8$ to $25$ simulation minutes (~16.6 min).
- *Rain Storm ($\times 3$ Inflow) with Open Valve:* Equilibrium settles at $H \approx 6.41\text{ m}$, staying safely below $8.0\text{ m}$.

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## C. Power Load Meter ( `power_load` )

- **Metric:** Substation electrical load in megawatts ($P$).
- **Operating Bounds:** $5.0 \le P \le 120.0\text{ MW}$.
- **Diurnal Target Demand ($P^(t)$):*

- **Evening Transition ($20:00 \le t < 21:00$):**
  $$P^*(t) = 30.0 + 14.0 \cdot (1.0 + \cos(\pi(t - 20.0)))$$
  Smooth cosine descent from $58.0\text{ MW}$ down to $30.0\text{ MW}$.
- **Deep Night ($21:00 \le t < 05:00$):**
  With $t_{\text{night}} = (t - 21.0) \pmod{24}$:
  $$P^*(t) = 30.0 + 3.0 \cdot \sin\left(\frac{2\pi \cdot t_{\text{night}}}{8.0}\right)$$
  Night load oscillates gently between $27.0\text{ MW}$ and $33.0\text{ MW}$ (strictly within $25 - 35\text{ MW}$).
- **Morning Transition ($05:00 \le t < 06:00$):**
  $$P^*(t) = 30.0 + 14.0 \cdot (1.0 - \cos(\pi(t - 5.0)))$$
  Smooth cosine ascent from $30.0\text{ MW}$ up to $58.0\text{ MW}$.
- **Daytime ($06:00 \le t \le 20:00$):**
  $$P^*(t) = 30.0 + 70.0 \cdot \left(0.6 \cdot \sin\left(\frac{\pi(t - 6.0)}{14.0}\right) + 0.4\right)$$
  Daytime peak reaches $100.0\text{ MW}$ at 13:00 (within $85 - 110\text{ MW}$).
- Seamless continuity at boundary hours $05:00$, $06:00$, $20:00$, and $21:00$ prevents jumps $> 3\text{ MW}$ between consecutive simulation minutes (max step $< 0.75\text{ MW/min}$).
- **Actuation Impact ( `switch_grid` ):**
- Grid state factors: `grid_a` ($1.0$), `grid_b` ($0.85$), `island` ($0.60$), `shed` ($0.40$), `standby` ($0.20$).
- **Incident Impact ( `grid_overload` ):**
- Target demand is set to $\max(P^*(t) \cdot 1.35, 118.0\text{ MW})$, ensuring the overload is visible and exceeds the $114.0\text{ MW}$ threshold at any hour of day or night.
- If true power exceeds $114.0\text{ MW}$ continuously for 60 simulation seconds, **blackout** triggers: power target drops to $10\%$ of normal target and emergency calls surge ($+4.0$).
- **Dynamics:** Time constant $\tau = 120\text{ s}$, noise $\sigma = 0.3\sqrt{\Delta t}\text{ MW}$. Steady-state spread $\sigma\sqrt{\tau/2} \approx 2.32\text{ MW}$.

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## D. Air Quality Station ( `air_quality` )

- **Metric:** Ambient Air Quality Index ($AQI$).
- **Operating Bounds:** $10.0 \le AQI \le 350.0$.
- **Target Formulation:**
  $$\text{Target} = \text{Background}(\text{zone}) + 0.45 \cdot \text{mean}(\text{zone true traffic}) + \text{IndustrialTerm} + \text{IncidentTerms}$$
- **Parameters:**
- Zone Backgrounds: `zone-1` ($18.0$), `zone-2` ($15.0$), `zone-3` ($20.0$), `zone-4` ($8.0$), `zone-5` ($14.0$), `zone-6` ($12.0$).
- Clean Baseline: In `zone-4` (Green Hills), background of $8.0$ yields measured AQI of about $19$ at night and about $60$ at rush hour.
- Industrial Contribution: `zone-3` receives $+25.0\text{ AQI}$.
- Incidents:
  - `fire` : $+150.0\text{ AQI}$ ramping linearly over 120 simulation seconds.
  - `air_hazard` : $+100.0\text{ AQI}$, mitigated by signal plans ( `red_restriction` reduces bonus by 50%, `all_red` reduces bonus by 80%).
- **Dynamics:** Time constant $\tau = 300\text{ s}$, noise $\sigma = 0.15\sqrt{\Delta t}\text{ AQI}$. Steady-state spread $\sigma\sqrt{\tau/2} \approx 1.84\text{ AQI}$.

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## E. Emergency Call Dispatch ( `emergency_call` )

- **Metric:** Municipal emergency 911/112 call volume in calls per minute ($C$).
- **Operating Bounds:** $0.0 \le C \le 30.0\text{ calls/min}$.
- **Arrival Process:**

- In each tick, arrivals $n \sim \text{Poisson}(\lambda \cdot \Delta t / 60)$ are drawn from `physics_rng`.
- Rate parameter $\lambda$ (calls/min) is dynamic per zone:
  $$\lambda = 2.0 + 4.5 \cdot \mathbb{I}(\text{any zone cam} > 120) + 8.0 \cdot \mathbb{I}(\text{any zone water} > 8.0) + \text{IncidentSurges} - 1.0 \cdot N_{\text{generic_dispatches}}$$
  Floored at $\lambda_{\text{min}} = 0.2\text{ calls/min}$.
- Incident surges: `traffic_accident` ($+6.0$), `fire` ($+8.0$), `grid_overload` with blackout ($+4.0$).
- **Telemetry Reporting:**
- Reported reading is the exact count of arrivals during the past 60 simulation seconds, maintained in a FIFO deque per zone and capped at 30 calls/min.
- Active generic dispatches relieve emergency pressure for `DISPATCH_DURATION_S` (default 900 s).

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## 3. Physical Incident Engine

Physical incidents represent real municipal emergencies, distinct from adversary cyber attacks. They alter physical reality directly and can be resolved through timely city actuation.

| Incident Type | Scope | Physical Effect | Resolution & Duration Defaults |
|---|---|---|---|
| `rain_storm` | Zone | Upstream inflow $Q_{\text{in}}$ multiplied by intensity ($1.0 - 5.0$, default $3.0$). | Defaults to `duration_s = 1800` (30 min) when omitted; naturally expires. |
| `traffic_accident` | Device (Camera) | Camera target multiplied by $\times 0.3$ ramped over 60 s; zone $\lambda + 6.0$. | Lasts until resolved by `dispatch` matching `traffic_accident`, `medical_emergency`, or `medical_surge` after 300 s response time (or cleared). |
| `fire` | Zone | AQI target $+150$ ramped over 120 s; zone $\lambda + 8.0$. | Lasts until resolved by `dispatch` matching `fire` after 180 s response time (or cleared). |
| `grid_overload` | Zone (Power) | Power target $\max(P^* \cdot 1.35, 118\text{ MW})$. Continuous load $> 114\text{ MW}$ for 60 s causes blackout (target $\times 0.10$, $\lambda + 4.0$). | Pre-blackout: resolved immediately by `switch_grid` to `grid_b`, `shed`, or `standby`. Post-blackout: resolved by `dispatch("blackout")` after 240 s. Lasts until resolved or cleared. |
| `air_hazard` | Zone | AQI target $+100$, mitigated by `red_restriction` or `all_red` signal plans. | Defaults to `duration_s = 1800` (30 min) when omitted; naturally expires. |

### Zero-Trust Isolation of Physical Incidents

- **MCP Invisibility:** Physical events are never exposed to City Brain or Guardian through MCP queries.
- **Contained / Attacked Commands:** Any command refused due to device containment ( `DEVICE_CONTAINED` ) or suppressed by an actuator attack does **not** resolve physical incidents.
- **Red Team API (Port 8100):** Protected by `X-Red-Team-Token` :
- `POST /api/event/inject` : Injects incident ( `type` , `zone_id` , `device_id` , `intensity` , `duration_s` ).
- `POST /api/event/clear` : Clears specific incident or all incidents.

- `GET /api/event/active`: Lists active physical events.
- **Ground Truth Ledger:** Initiations, durations, and resolutions are recorded in SQLite table `physical_events`.

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## 4. Adversary Capability Engine

The adversary model corrupts wire telemetry while the internal SQLite ledger preserves ground truth:

| Attack Capability | Wire Manipulation Method | Detection & Guardian Mitigation |
|---|---|---|
| **`device_physical`** | Overwrites telemetry value with attacker-defined spoofing; generates valid Ed25519 signature with legitimate device key. | Guardian flags anomalous physical gradients or cross-sensor contradictions; invokes `rollback_reading` or `isolate_device`. |
| **`device_key`** | Attacker extracts valid private key and broadcasts malicious readings that cryptographically verify on wire. | Guardian cross-checks with historical sensor baselines and neighbor correlations; invokes `quarantine_device`. |
| **`link`** | Flips payload bits (`corrupt_bytes`), drops readings (`drop`), or replays past packets (`replay`). | Guardian detects invalid Ed25519 signatures, sequence gaps, or timestamp regression; alerts operators. |
| **`rogue`** | Broadcasts telemetry under an unregistered identifier (e.g., `rg-01`). | Guardian compares against `list_devices`; quaranines rogue hardware via MCP `quarantine_device`. |
| **`actuator`** | Acknowledges MCP commands as `"ok"` but leaves true physical state unchanged. | City Brain observes telemetry failing to converge toward expected physical setpoints. |
| **`sensor_fault`** | Injects subtle linear drift ($\Delta t \cdot \text{drift_rate}$), stuck frozen values, or zero dropouts. | Statistical moving-average and deviation filters identify sensor drift. |